National Testing Agency

Question Paper Name :	B Tech 2 Apr 2025 Shift 1
Subject Name :	B. Tech
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A = \begin{bmatrix} \alpha & -1 \\ 6 & \beta \end{bmatrix}, \alpha > 0$, such that $\det(A) = 0$ and $\alpha + \beta = 1$. If I denotes 2×2 identity matrix, then the matrix $(I + A)^8$ is :

Options :

A. $\begin{bmatrix} 257 & -64 \\ 514 & -127 \end{bmatrix}$

B. $\begin{bmatrix} 766 & -255 \\ 1530 & -509 \end{bmatrix}$

C. $\begin{bmatrix} 1025 & -511 \\ 2024 & -1024 \end{bmatrix}$

D. $\begin{bmatrix} 4 & -1 \\ 6 & -1 \end{bmatrix}$

Question Number : 2 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, QR $= 5$ and the coordinates of the point P be (0,2,3). If the area of the triangle PQR is $\frac{m}{n}$ then :

Options :

A. $2\,m - 5\sqrt{21}\,n = 0$

B. $m - 5\sqrt{21}\,n = 0$

C. $5\,m - 21\sqrt{2}\,n = 0$

D. $5\,m - 2\sqrt{21}\,n = 0$

Question Number : 3 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $a \in R$ and A be a matrix of order 3×3 such that $\det(A) = -4$ and $A + I = \begin{bmatrix} 1 & a & 1 \\ 2 & 1 & 0 \\ a & 1 & 2 \end{bmatrix}$, where I is the identity matrix of order 3×3. If $\det((a+1)\text{adj}((a-1)A))$ is $2^m 3^n$, m, n $\in \{0,1,2, \ldots, 20\}$, then m + n is equal to :

Options :

A. 14

B. 17

C. 15

D. 16

Question Number : 4 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2}\,\cos^2\theta + (2 - \sqrt{6}\,)\cos\theta - \sqrt{3}\, = 0$, is equal to:

Options :

A. 8

B. 6

C. 10

D. 12

Question Number : 5 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If S and S' are the foci of the ellipse $\frac{x^2}{18}+\frac{y^2}{9}=1$ and P be a point on the ellipse, then $\min\left(SP\cdot S'P\right)+\max\left(SP\cdot S'P\right)$ is equal to :

Options :

A. $3(6+\sqrt{2}\ )$

B. $3(1+\sqrt{2}\ )$

C. 27

D. 9

Question Number : 6 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the focal chord PQ of the parabola $y^2=4x$ make an angle of $60°$ with the positive x axis, where P lies in the first quadrant. If the circle, whose one diameter is PS, S being the focus of the parabola, touches the y-axis at the point $(0,\alpha)$, then $5\alpha^2$ is equal to:

Options :

A. 15

B. 25

C. 20

D. 30

Question Number : 7 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the function $f(x)=2x^3-9ax^2+12a^2x+1$, where $a>0$, attains its local maximum and local minimum values at p and q , respectively, such that $p^2=q$, then $f(3)$ is equal to :

Options :

A. 55

B. 37

C. 10

D. 23

Question Number : 8 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $ABCD$ be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5,6 and 7 square units respectively. Then the area (in square units) of the $\triangle BCD$ is equal to :

Options :

A. $\sqrt{110}$

B. 12

C. $\sqrt{340}$

D. $7\sqrt{3}$

Question Number : 9 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let z be a complex number such that $|z| = 1$. If $\frac{2+k^2z}{k+\overline{z}} = kz, k \in \mathbf{R}$, then the maximum distance of $k + ik^2$ from the circle $|z - (1 + 2i)| = 1$ is :

Options :

A. $\sqrt{5} + 1$

B. 3

C. $\sqrt{3} + 1$

D. 2

Question Number : 10 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let one focus of the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be at $(\sqrt{10}, 0)$ and the corresponding directrix be $x = \frac{9}{\sqrt{10}}$. If e and l respectively are the eccentricity and the length of the latus rectum of H , then $9(e^2 + l)$ is equal to :

Options :

A. 12

B. 14

C. 15

D. 16

Question Number : 11 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The largest $n \in \mathbf{N}$ such that 3^n divides 50 ! is :

Options :

A. 22

B. 20

C. 21

D. 23

Question Number : 12 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $P_n = \alpha^n + \beta^n, n \in N$. If $P_{10} = 123, P_9 = 76, P_8 = 47$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ is :

Options :

A. $x^2 + x - 1 = 0$

B. $x^2 - x + 1 = 0$

C. $x^2 + x + 1 = 0$

D. $x^2 - x - 1 = 0$

Question Number : 13 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\vec{a}$ is a nonzero vector such that its projections on the vectors $2\hat{i} - \hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} - 2\hat{k}$ and $\hat{k}$ are equal, then a unit vector along $\vec{a}$ is :

Options :

A. $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} + 5\hat{k})$

B. $\frac{1}{\sqrt{155}}(-7\hat{i} + 9\hat{j} - 5\hat{k})$

C. $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} - 5\hat{k})$

D. $\frac{1}{\sqrt{155}}(7\hat{i} + 9\hat{j} + 5\hat{k})$

Question Number : 14 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the system of linear equations

$$3x + y + \beta z = 3$$
$$2x + \alpha y - z = -3$$
$$x + 2y + z = 4$$

has infinitely many solutions, then the value of $22\beta - 9\alpha$ is :

Options :

A. 31

B. 37

C. 43

D. 49

Question Number : 15 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f : \mathbf{R} \to \mathbf{R}$ be a twice differentiable function such that $(\sin x \cos y)(f(2x + 2y) - f(2x - 2y)) = (\cos x \sin y)(f(2x + 2y) + f(2x - 2y))$, for all $x, y \in \mathbf{R}$. If $f'(0) = \frac{1}{2}$, then the value of $24f''\left(\frac{5\pi}{3}\right)$ is :

Options :

A. 2

B. 3

C. -3

D. -2

Question Number : 16 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let A be the set of all functions $f : \mathbf{Z} \to \mathbf{Z}$ and R be a relation on A such that R = {(f, g) : $f(0) = g(1)$ and $f(1) = g(0)$}. Then R is :

Options :

A. Symmetric and transitive but not reflective

B. Symmetric but neither reflective nor transitive

C. Transitive but neither reflexive nor symmetric

D. Reflexive but neither symmetric nor transitive

Question Number : 17 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of sequences of ten terms, whose terms are either 0 or 1 or 2 , that contain exactly five 1 s and exactly three 2 s , is equal to :

Options :

A. 360

B. 2520

C. 1820

D. 45

Question Number : 18 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For $\alpha, \beta, \gamma \in \mathbf{R}$, if $\lim\limits_{x \to 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$, then $\beta + \gamma - \alpha$ is equal to :

Options :

A. -1

B. 4

C. 6

D. 7

Question Number : 19 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The term independent of x in the expansion of $\left(\frac{(x+1)}{(x^{2/3}+1-x^{1/3})} - \frac{(x-1)}{(x-x^{1/2})}\right)^{10}$, $x > 1$, is :

Options :

A. 240

B. 120

C. 150

D. 210

Question Number : 20 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $a_1, a_2, a_3, \ldots$ be in an A.P. such that $\sum_{k=1}^{12} a_{2k-1} = -\frac{72}{5}a_1, a_1 \neq 0$. If $\sum_{k=1}^{n} a_k = 0$, then n is :

Options :

A. 18

B. 17

C. 11

D. 10

Mathematics Section B

Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Three distinct numbers are selected randomly from the set $\{1,2,3, \ldots, 40\}$. If the probability, that the selected numbers are in an increasing G.P., is $\frac{m}{n}, \gcd(m,n) = 1$, then $m+n$ is equal to _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 22 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $f : \mathbf{R} \to \mathbf{R}$ be a thrice differentiable odd function satisfying $f'(x) \geq 0, f''(x) = f(x), f(0) = 0, f'(0) = 3$. Then $9f\left(\log_e 3\right)$ is equal to _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 23 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the area of the region $\left\{(x,y) : |4 - x^2| \leq y \leq x^2, y \leq 4, x \geq 0\right\}$ is $\left(\frac{80\sqrt{2}}{\alpha} - \beta\right), \alpha, \beta \in \mathbf{N}$, then $\alpha + \beta$ is equal to _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText

Question Number : 24 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The absolute difference between the squares of the radii of the two circles passing through the point $(-9,4)$ and touching the lines $x+y=3$ and $x-y=3$, is equal to _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 25 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let [.] denote the greatest integer function. If $\int_0^{e^3}\left[\frac{1}{e^{x-1}}\right]dx=\alpha-\log_e 2$, then α^3 is equal to _____.

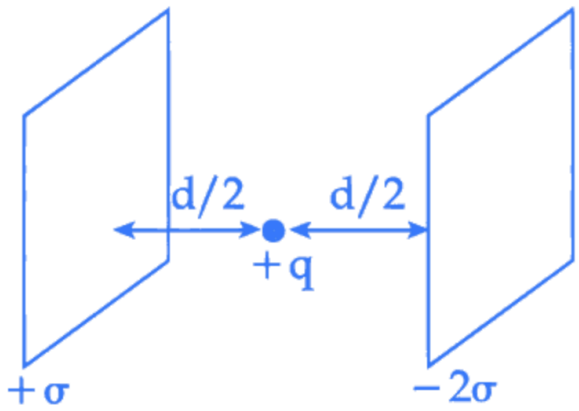
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Physics Section A

Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider two infinitely large plane parallel conducting plates as shown below. The plates are uniformly charged with a surface charge density $+\sigma$ and -2σ. The force experienced by a point charge +q placed at the mid point between two plates will be:



Options :

A. $\frac{3\sigma q}{4\epsilon_0}$

B. $\frac{3\sigma q}{2\epsilon_0}$

C. $\frac{\sigma q}{4\epsilon_0}$

D. $\frac{\sigma q}{2\epsilon_0}$

Question Number : 27 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A monochromatic light is incident on a metallic plate having work function ϕ. An electron, emitted normally to the plate from a point A with maximum kinetic energy, enters a constant magnetic field, perpendicular to the initial velocity of electron. The electron passes through a curve and hits back the plate at a point *B*. The distance between *A* and *B* is : (Given : The magnitude of charge of an electron is e and mass is m, h is Planck's constant and c is velocity of light. Take the magnetic field exists throughout the path of electron)

Options :

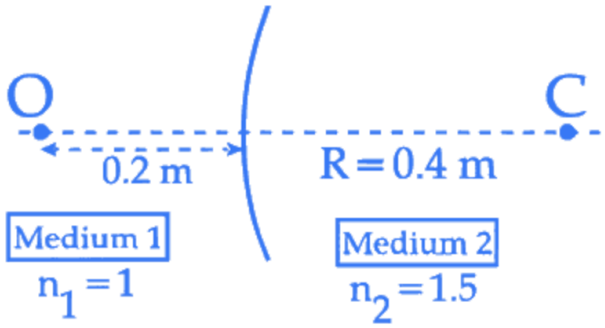
A. $\sqrt{2m\left(\frac{hc}{\lambda}-\phi\right)}/eB$

B. $\sqrt{8\,m\left(\frac{hc}{\lambda}-\phi\right)}/eB$

C. $\sqrt{m(hc/\lambda-\phi)}\ /eB$

D. $2\sqrt{m(hc/\lambda-\phi)}\ /eB$

Question Number : 28 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical



A spherical surface separates two media of refractive indices 1 and 1.5 as shown in figure. Distance of the image of an object ' O ', is :

(C is the center of curvature of the spherical surface and R is the radius of curvature)

Options :

A. 0.24 m left to the spherical surface

B. 0.24 m right to the spherical surface

C. 0.4 m left to the spherical surface

D. 0.4 m right to the spherical surface

Question Number : 29 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A river is flowing from west to east direction with speed of 9 km h^{-1}. If a boat capable of moving at a maximum speed of 27 km h^{-1} in still water, crosses the river in half a minute, while moving with maximum speed at an angle of 150° to direction of river flow, then the width of the river is :

Options :

A. 112.5 m

B. 75 m

C. 300 m

D. $112.5 \times \sqrt{3}$ m

Question Number : 30 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Moment of inertia of a rod of mass ' M ' and length ' L ' about an axis passing through its center and normal to its length is ' α '. Now the rod is cut into two equal parts and these parts are joined symmetrically to form a cross shape. Moment of inertia of cross about an axis passing through its center and normal to plane containing cross is :

Options :

A. $\alpha/4$

B. $\alpha/8$

C. α

D. $\alpha/2$

Question Number : 31 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let B_1 be the magnitude of magnetic field at center of a circular coil of radius R carrying current I. Let B_2 be the magnitude of magnetic field at an axial distance ' x ' from the center. For x : R = 3 : 4, $\frac{B_2}{B_1}$ is :

Options :

A. 64 : 125

B. 25 : 16

C. $4:5$

D. $16:25$

Question Number : 32 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A light wave is propagating with plane wave fronts of the type $x + y + z =$ constant. Th angle made by the direction of wave propagation with the x-axis is :

Options :

A. $\cos^{-1}(2/3)$

B. $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

C. $\cos^{-1}\left(\frac{1}{3}\right)$

D. $\cos^{-1}\left(\sqrt{\frac{2}{3}}\right)$

Question Number : 33 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match List I with List II.

	List - I		**List - II**
(A)	Coefficient of viscosity	(I)	$\left[ML^0 T^{-3}\right]$
(B)	Intensity of wave	(II)	$\left[ML^{-2} T^{-2}\right]$
(C)	Pressure gradient	(III)	$\left[M^{-1}LT^2\right]$
(D)	Compressibility	(IV)	$\left[ML^{-1} T^{-1}\right]$

Choose the correct answer from the options given below:

Options :

A. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

B. (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

Question Number : 34 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A particle is subjected to two simple harmonic motions as :

$$x_1 = \sqrt{7}\ \sin 5t\text{cm}$$

and $x_2 = 2\sqrt{7}\ \sin\left(5t + \frac{\pi}{3}\right)$cm where x is displacement and t is time in seconds. The maximum acceleration of the particle is $x \times 10^{-2}$ ms^{-2}. The value of x is :

Options :

A. $5\sqrt{7}$

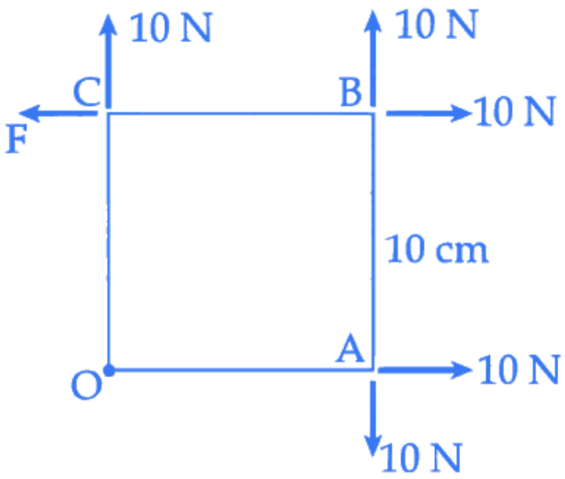
B. 125

C. $25\sqrt{7}$

D. 175

Question Number : 35 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A square Lamina OABC of length 10 cm is pivoted at ' O'. Forces act at Lamina as shown in figure. If Lamina remains stationary, then the magnitude of F is :



Options :

A. 10 N

B. 0 (zero)

C. $10\sqrt{2}$ N

D. 20 N

Question Number : 36 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0,0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is:

Options :

A. $(3d/4,0,0)$

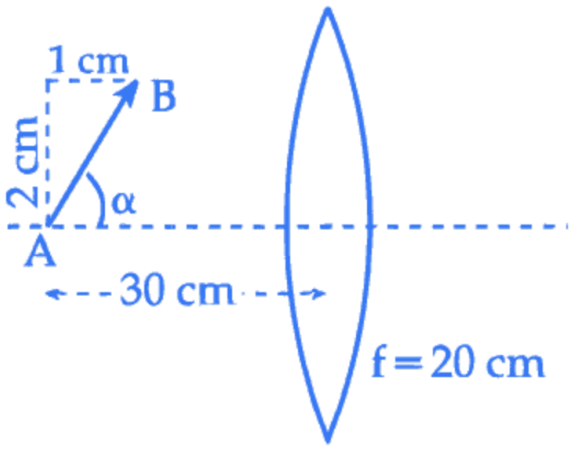
B. $(d/4,0,0)$

C. $(4d/3,0,0)$

D. $(d/3,0,0)$

Question Number : 37 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A slanted object AB is placed on one side of convex lens as shown in the diagram. The image is formed on the opposite side. Angle made by the image with principal axis is :



Options :

A. $-\alpha$

B. $-\frac{\alpha}{2}$

C. -45°

D. $+45^{\circ}$

Question Number : 38 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Considering Bohr's atomic model for hydrogen atom :

(A) the energy of H atom in ground state is same as energy of He^{+}ion in its first excited state.

(B) the energy of H atom in ground state is same as that for Li^{++} ion in its second excited state.

(C) the energy of H atom in its ground state is same as that of He^{+}ion for its ground state.

(D) the energy of He^{+}ion in its first excited state is same as that for Li^{++}ion in its ground state.

Choose the correct answer from the options given below :

Options :

A. (A), (B) only

B. (A), (D) only

C. (A), (C) only

D. (B), (D) only

Question Number : 39 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In an adiabatic process, which of the following statements is true?

Options :

A. The internal energy of the gas decreases as the temperature increases

B. The molar heat capacity is zero

C. Work done by the gas equals the increase in internal energy

D. The molar heat capacity is infinite

Question Number : 40 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The equation for real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where P, V, T and R are the pressure, volume, temperature and gas constant, respectively. The dimension of ab^{-2} is equivalent to that of :

Options :

A. Compressibility

B. Planck's constant

C. Energy density

D. Strain

Question Number : 41 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The relationship between the magnetic susceptibility (x) and the magnetic permeability (μ) is given by :

(μ_0 is the permeability of free space and μ_T is relative permeability)

Options :

A. $X = \frac{\mu}{\mu_0} - 1$

B. $X = \frac{\mu_r}{\mu_0} + 1$

C. $\chi = \mu_r + 1$

D. $X = 1 - \frac{\mu}{\mu_0}$

Question Number : 42 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The battery of a mobile phone is rated as 4.2 V, 5800 mAh. How much energy is stored ir it when fully charged?

Options :

A. 87.7 kJ

B. 43.8 kJ

C. 48.7 kJ

D. 24.4 kJ

Question Number : 43 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A zener diode with 5 V zener voltage is used to regulate an unregulated dc voltage input of 25 V . For a 400Ω resistor connected in series, the zener current is found to be 4 times load current. The load current (I_L) and load resistance (R_L) are :

Options :

A. $I_L = 0.02\ mA; R_L = 250\Omega$

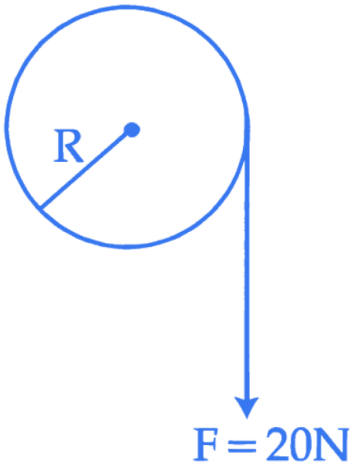
B. $I_L = 10\ A; R_L = 0.5\Omega$

C. $I_L = 10\ mA; R_L = 500\Omega$

D. $I_L = 20\ mA; R_L = 250\Omega$

Question Number : 44 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A cord of negligible mass is wound around the rim of a wheel supported by spokes with negligible mass. The mass of wheel is 10 kg and radius is 10 cm and it can freely rotate without any friction. Initially the wheel is at rest. If a steady pull of 20 N is applied on the cord, the angular velocity of the wheel, after the cord is unwound by 1 m , would be:



Options :

A. 20 rad/s

B. 30 rad/s

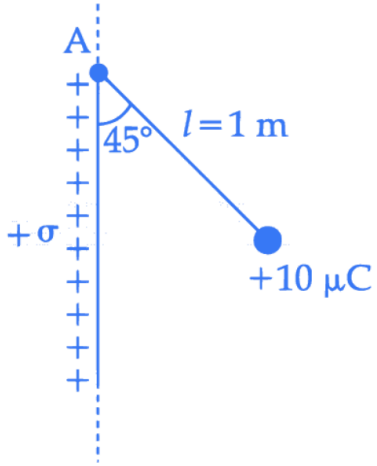
C. 10 rad/s

D. 0 rad/s

Question Number : 45 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A small bob of mass 100 mg and charge $+10\mu C$ is connected to an insulating string of length 1 m . It is brought near to an infinitely long non-conducting sheet of charge density ' σ ' as shown in figure. If string subtends an angle of 45° with the sheet at equilibrium the charge density of sheet will be.

(Given, $\epsilon_0 = 8.85 \times 10^{-12} \frac{F}{m}$ and acceleration due to gravity, $g = 10 \frac{m}{s^2}$)



Options :

A. $1.77\ \text{nC/m}^2$

B. $0.885\ \text{nC/m}^2$

C. $885\ \text{nC/m}^2$

D. $17.7\ \text{nC/m}^2$

Physics Section B

Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A steel wire of length 2 m and Young's modulus $2.0 \times 10^{11}\ \text{N m}^{-2}$ is stretched by a force. If Poisson ratio and transverse strain for the wire are 0.2 and 10^{-3} respectively, then the elastic potential energy density of the wire is _____ $\times 10^5$ (in SI units).

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 47 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

γ_A is the specific heat ratio of monoatomic gas A having 3 translational degrees of freedom. γ_B is the specific heat ratio of polyatomic gas B having 3 translational, 3 rotational degrees of freedom and 1 vibrational mode. If $\frac{\gamma_A}{\gamma_B} = \left(1 + \frac{1}{n}\right)$, then the value of n is _____ .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 48 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A vessel with square cross-section and height of 6 m is vertically partitioned. A small window of $100\ \text{cm}^2$ with hinged door is fitted at a depth of 3 m in the partition wall. One part of the vessel is filled completely with water and the other side is filled with the liquid having density $1.5 \times 10^3\ \text{kg/m}^3$. What force one needs to apply on the hinged door so that it does not get opened ?

$$(\text{Acceleration due to gravity } = 10\ \text{m/s}^2)$$

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 49 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the measured angular separation between the second minimum to the left of the central maximum and the third minimum to the right of the central maximum is 30° in a single slit diffraction pattern recorded using 628 nm light, then the width of the slit is _____ μm.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 50 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A person travelling on a straight line moves with a uniform velocity v_1 for a distance x and with a uniform velocity v_2 for the next $\frac{3}{2}x$ distance. The average velocity in this motion is $\frac{50}{7}$ m/s. If v_1 is 5 m/s then v_2 = _____ m/s.

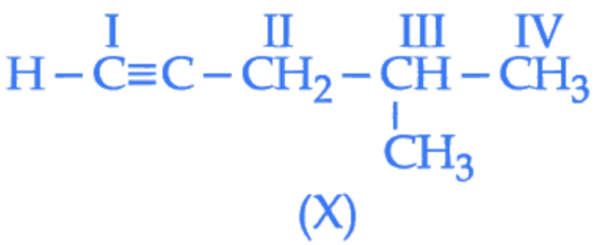
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Chemistry Section A

Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following compound (X)



(X)

The most stable and least stable carbon radicals, respectively, produced by homolytic cleavage of corresponding C – H bond are :

Options :

A. III, II

B. II, IV

C. I, IV

D. II, I

Question Number : 52 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

Consider the above reaction, what mass of $CaCl_2$ will be formed if 250 mL of 0.76 M HCl reacts with 1000 g of $CaCO_3$?

(Given : Molar mass of Ca, C, O, H and Cl are 40,12,16,1 and 35.5 g mol^{-1}, respectively)

Options :

A. 3.908 g

B. 2.636 g

C. 10.545 g

D. 5.272 g

Question Number : 53 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If equal volumes of AB_2 and XY (both are salts) aqueous solutions are mixed, which of the following combination will give a precipitate of AY_2 at 300 K ? (Given K_{sp} (at 300 K) for $AY_2 = 5.2 \times 10^{-7}$)

Options :

A. $2.0 \times 10^{-4} MAB_2, 0.8 \times 10^{-3} MXY$

B. $2.0 \times 10^{-2} MAB_2, 2.0 \times 10^{-2} MXY$

C. $1.5 \times 10^{-4} MAB_2, 1.5 \times 10^{-3} MXY$

D. $3.6 \times 10^{-3} MAB_2, 5.0 \times 10^{-4} MXY$

Question Number : 54 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : The metallic radius of Al is less than that of Ga .

Statement (II) : The ionic radius of Al^{3+} is less than that of Ga^{3+}.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options :

A. Statement I is correct but Statement II is incorrect

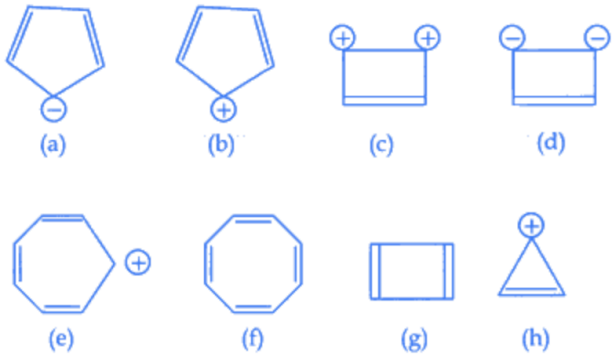
B. Both Statement I and Statement II are correct

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are incorrect

Question Number : 55 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Designate whether each of the following compounds is aromatic or not aromatic.



Options :

A. a, c, d, e, h aromatic and b, f, g not aromatic

B. b, e, f, g aromatic and a, c, d, h not aromatic

C. a, b, c, d aromatic and e, f, g, h not aromatic

D. e, g aromatic and a, b, c, d, f, h not aromatic

Question Number : 56 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Choose the correct tests with respective observations.

(A) $CuSO_4$ (acidified with acetic acid) $+K_4\left[Fe(CN)_6\right] \rightarrow$ Chocolate brown precipitate.

(B) $FeCl_3 + K_4\left[Fe(CN)_6\right] \rightarrow$ Prussian blue precipitate.

(C) $ZnCl_2 + K_4\left[Fe(CN)_6\right]$, neutralised with $NH_4OH \rightarrow$ White or bluish white precipitate.

(D) $MgCl_2 + K_4\left[Fe(CN)_6\right] \rightarrow$ Blue precipitate.

(E) $BaCl_2 + K_4\left[Fe(CN)_6\right]$, neutralised with $NaOH \rightarrow$ White precipitate.

Choose the correct answer from the options given below :

Options :

A. A, D and E only

B. A, B and C only

C. B, D and E only

D. C, D and E only

Question Number : 57 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A solution is made by mixing one mole of volatile liquid *A* with 3 moles of volatile liquid *B*. The vapour pressure of pure A is 200 mm Hg and that of the solution is 500 mm Hg . The vapour pressure of pure B and the least volatile component of the solution, respectively, are:

Options :

A. 1400 mm Hg, A

B. 1400 mm Hg, *B*

C. 600 mm Hg, A

D. 600 mm Hg, B

Question Number : 58 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A molecule with the formula AX_4Y has all it's elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y . Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is:

Options :

A. Pentagonal planar

B. Square pyramidal

C. Trigonal bipyramidal

D. Octahedral

Question Number : 59 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : In octahedral complexes, when $\Delta_0 <$ P high spin complexes are formed. When $\Delta_0 > P$ low spin complexes are formed.

Statement (II) : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options :

A. Both Statement I and Statement II are incorrect

B. Statement I is correct but Statement II is incorrect

C. Statement I is incorrect but Statement II is correct

D. Both Statement I and Statement II are correct

Question Number : 60 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

On complete combustion 1.0 g of an organic compound (X) gave 1.46 g of CO_2 and 0.567 g of H_2O. The empirical formula mass of compound (X) is _____ g. (Given molar mass in $gmol^{-1}$ C : 12, H : 1, O : 16)

Options :

A. 60

B. 45

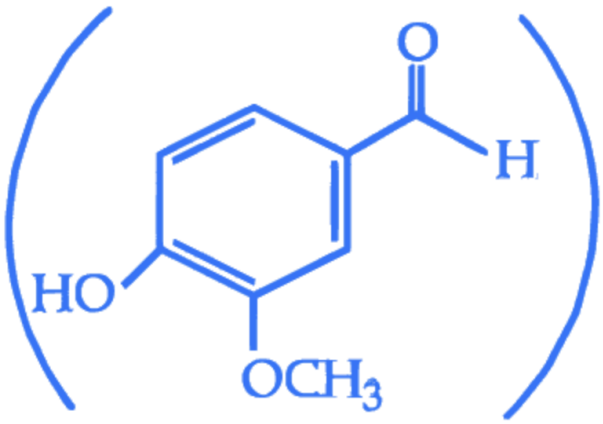
C. 30

D. 15

Question Number : 61 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

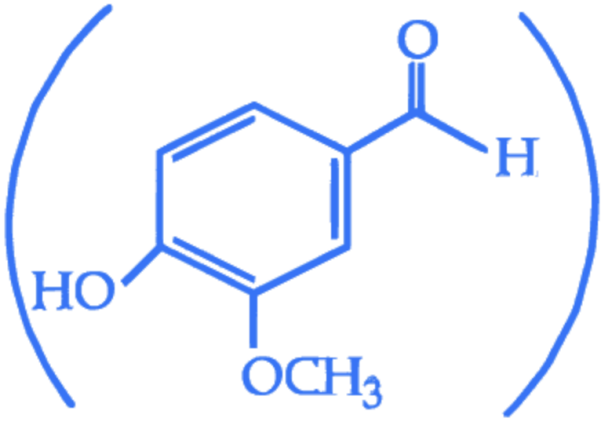
Given below are two statements:

Statement I : Vanilin



will react with NaOH and also with Tollen's reagent.

Statement II : Vanilin



will undergo self aldol condensation very easily.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options :

A. Statement I is correct but Statement II is incorrect

B. Both Statement I and Statement II are correct

C. Both Statement I and Statement II are incorrect

D. Statement I is incorrect but Statement II is correct

Question Number : 62 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An optically active alkyl halide C_4H_9Br[A] reacts with hot KOH dissolved in ethanol and forms alkene [*B*] as major product which reacts with bromine to give dibromide [*C*]. The compound [C] is converted into a gas [D] upon reacting with alcoholic $NaNH_2$. During hydration 18 gram of water is added to 1 mole of gas [D] on warming with mercuric sulphate and dilute acid at 333 K to form compound [E]. The IUPAC name of compound [E] is :

Options :

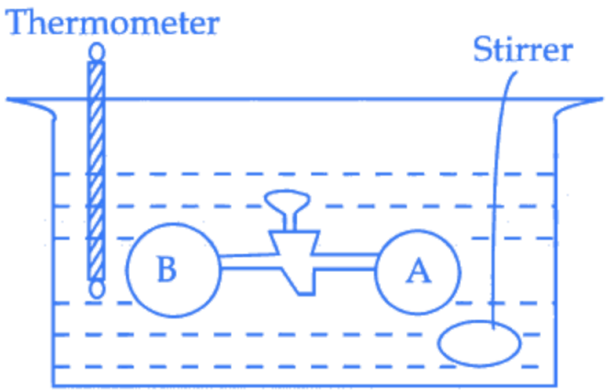
A. Butan-2-ol

B. But-2-yne

C. Butan-2-one

D. Butan-1-al

Question Number : 63 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical



Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true ?

Options :

A. $dq \neq 0$

B. The pressure in the vessel B before opening the stopcock is zero

C. $dw \neq 0$

D. $dU \neq 0$

Question Number : 64 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Among $SO_2, NF_3, NH_3, XeF_2, ClF_3$ and SF_4, the hybridization of the molecule with nonzero dipole moment and highest number of lone-pairs of electrons on the central atom is

Options :

A. sp^3

B. dsp^2

C. $sp^3 d^2$

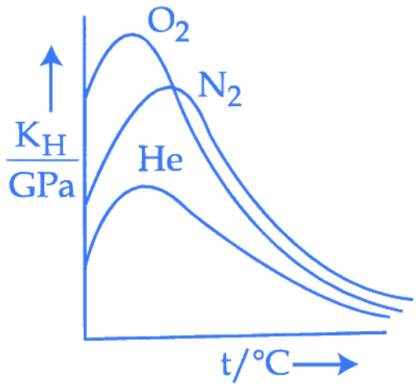
D. $sp^3 d$

Question Number : 65 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

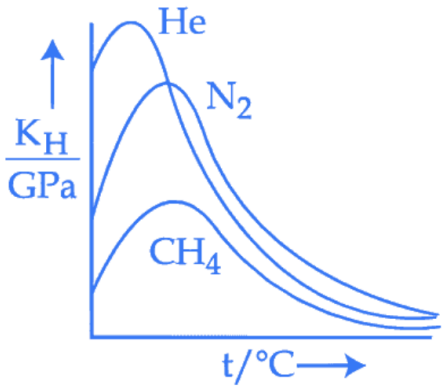
Which of the following graph correctly represents the plots of K_H at 1 bar for gases in water versus temperature?

Options :

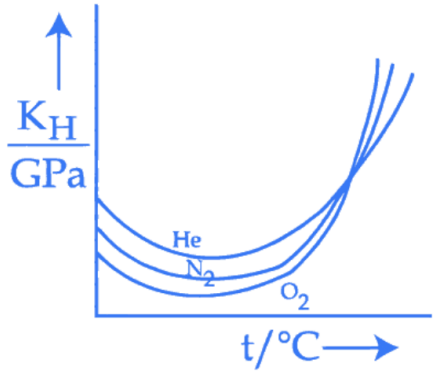
A.



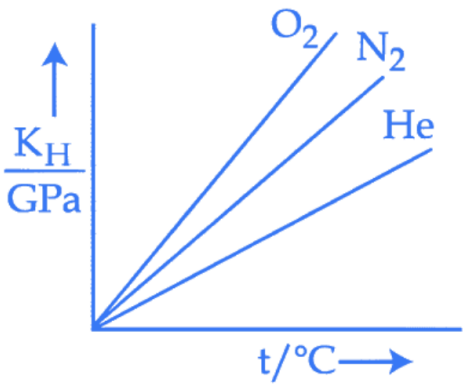
B.



C.



D.



Question Number : 66 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

According to Bohr's model of hydrogen atom, which of the following statement is incorrect?

Options :

A. Radius of 4th orbit is four times larger than that of 2nd orbit

B. Radius of 8th orbit is four times larger than that of 4th orbit

C. Radius of 6th orbit is three times larger than that of 4th orbit

D. Radius of 3rd orbit is nine times larger than that of 1st orbit

Question Number : 67 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the correct statement among the following :

Options :

A. All naturally occurring amino acids except glycine contain one chiral centre.

B. Amino acid, cysteine can easily undergo dimerisation due to the presence of free SH group.

C. Glutamic acid is the only amino acid that contains a $-COOH$ group at the side chain.

D. All naturally occurring amino acids are optically active.

Question Number : 68 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The correct order of basic nature in aqueous solution for the bases

$NH_3, H_2N-NH_2, CH_3CH_2NH_2, (CH_3CH_2)_2NH$ and $(CH_3CH_2)_3N$ is :

Options :

A. $NH_3 < H_2N-NH_2 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$

B. $H_2N-NH_2 < NH_3 < (CH_3CH_2)_3N < CH_3CH_2NH_2 < (CH_3CH_2)_2NH$

C. $NH_2-NH_2 < NH_3 < CH_3CH_2NH_2 < (CH_3CH_2)_3N < (CH_3CH_2)_2NH$

D. $NH_3 < H_2N-NH_2 < CH_3CH_2NH_2 < (CH_3CH_2)_2NH < (CH_3CH_2)_3N$

Question Number : 69 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The property/properties that show irregularity in first four elements of group-17 is/are

(A) Covalent radius

(B) Electron affinity

(C) Ionic radius

(D) First ionization energy

Choose the correct answer from the options given below :

Options :

A. A, B, C and D

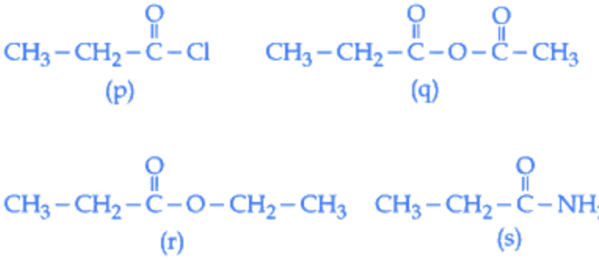
B. A and C only

C. B only

D. B and D only

Question Number : 70 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following molecules:



The correct order of rate of hydrolysis is :

Options :

A. $p > q > r > s$

B. $r > q > p > s$

C. $p > r > q > s$

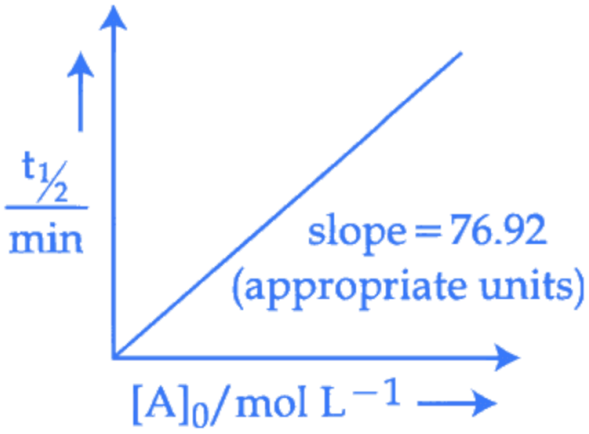
D. $q > p > r > s$

Chemistry Section B

Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For the reaction A → products.



The concentration of A at 10 minutes is _____ $\times 10^{-3}$ mol L^{-1} (nearest integer). The reaction was started with 2.5 mol L^{-1} of A .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 72 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

0.1 mol of the following given antiviral compound (P) will weigh _____ $\times 10^{-1}$ g (nearest integer).



(P)

(Given : molar mass in $gmol^{-1}$ H : 1, C : 12, N : 14, O : 16, F : 19, I : 127)

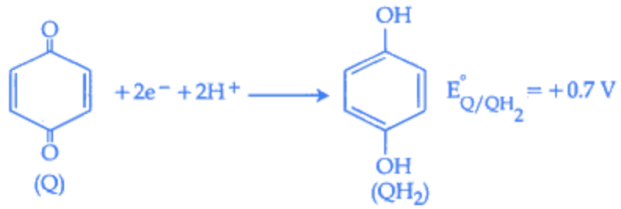
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 73 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following electrochemical cell at standard condition.

$$Au(s)|QH_2, Q|NH_4X(0.01M)||Ag^+(1M)\,|\,Ag(s) \quad E_{cell} = +0.4\ V$$

The couple QH_2/Q represents quinhydrone electrode, the half cell reaction is given below:



$$\left[\text{Given : } E^0_{Ag^+/Ag} = +0.8\ V \text{ and } \frac{2.303RT}{F} = 0.06\ V\right]$$

The pK_b value of the ammonium halide salt (NH_4X) used here is _____ . (nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 74 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Consider the following equilibrium,

$$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$$

0.1 mol of CO along with a catalyst is present in a $2dm^3$ flask maintained at 500 K . Hydrogen is introduced into the flask until the pressure is 5 bar and 0.04 mol of CH_3OH is formed. The K_p^θ is _____ $\times 10^{-3}$ (nearest integer).

Given : $R = 0.08\ dm^3\ bar\ K^{-1}\ mol^{-1}$

Assume only methanol is formed as the product and the system follows ideal gas behaviour.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 75 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A transition metal (M) among Mn, Cr, Co and Fe has the highest standard electrode potential (M^{3+}/M^{2+}). It forms a metal complex of the type $[M(CN)_6]^{4-}$. The number of electrons present in the e_g orbital of the complex is _____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText